

Test-Only Evaluation of Py-Feat Detector_{v2} on AFLFP Landmark Localization and DISFA Action Units

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Abstract—Pretrained facial behavior toolkits are frequently used outside the datasets on which their components were developed, yet deployment-oriented, test-only measurements are often unavailable for specialized populations. We evaluate the Py-Feat Detector_{v2} pipeline without fitting, fine-tuning, calibration, or threshold selection on two complementary benchmarks. The AFLFP cohort contains 1,136 facial-landmark images, balanced across 88 subjects and 16 movements; landmark localization reaches a 100.0% detection rate and 6.368% mean normalized mean error (NME). The DISFA cohort contains 5,400 spontaneous left-camera frames, uniformly balanced at 200 frames for each of 27 subjects; action-unit detection reaches 100.0% detection, macro F1 0.714, and macro ICC(3,1) 0.467. Inference throughput is 24.96 frames/s on AFLFP and 26.14 frames/s on DISFA on an Apple M1 Pro using Metal Performance Shaders. AFLFP error is stable across movements, whereas DISFA performance is heterogeneous: AU25 achieves F1 0.959 and AU20 achieves 0.479. The results establish a reproducible external baseline for Py-Feat, while exposing class imbalance, intensity-scale mismatch, and sampling limits that preclude claims of state-of-the-art superiority.

Index Terms—action units, AFLFP, DISFA, facial landmarks, facial palsy, Py-Feat, test-only evaluation

I. INTRODUCTION

Facial behavior pipelines combine face detection, geometric localization, and expression inference behind a convenient interface. Py-Feat is an open-source toolbox designed to make these components accessible and benchmarkable [1]. A package-level benchmark, however, does not directly answer a deployment question: how does the installed pretrained pipeline behave on a specialized dataset when no task-specific fitting is permitted?

We study that question in two different regimes. AFLFP was designed around facial palsy and contains asymmetric expressions with 68-point manual landmarks [2]. DISFA contains spontaneous video with frame-level intensities for 12 Facial Action Coding System action units (AUs) [3]. The former evaluates geometry under atypical facial motion; the latter evaluates expression scores under natural temporal and class imbalance.

This report makes three contributions. First, it gives an explicitly *test-only* measurement of Py-Feat Detector_{v2}: pretrained weights are frozen and no parameter, threshold, or sample is selected using benchmark performance. Second, deterministic cohorts cover every subject and target class while limiting the computation required by the full video corpus. Third, aggregate JSON, sample-level CSV, tables, and

vector figures form one auditable chain. Every number in the manuscript is regenerated from the saved predictions.

The study is an external evaluation, not a new model and not a controlled leaderboard entry. In particular, “test-only” describes the actions performed in this study; it does not certify that every upstream pretraining source is disjoint from AFLFP or DISFA.

II. BENCHMARKS AND PRIOR EVIDENCE

A. Py-Feat

Py-Feat provides a common interface for facial feature detection, preprocessing, analysis, and visualization [1]. The evaluated installation is version 2.0.3, using the packaged Detector_{v2} pipeline. We treat the complete pipeline as the object of evaluation and do not attribute performance to any single internal subcomponent.

B. AFLFP

AFLFP was introduced for facial landmark localization in facial palsy. It contains 88 subjects, 16 instructed facial movements, and independent manual annotations of 68 landmarks [2]. The released local distribution inspected here contains 6,476 JPEG files and 5,328 paired manual landmark annotations. The dataset paper reports NME normalized by the square root of ground-truth face-box area and a mean of 11.6% for its Cascaded FCN baseline [2]. We retain that published row as descriptive context, but its training protocol and evaluated image set are not identical to ours.

C. DISFA

DISFA records 27 adults viewing emotion-eliciting video and supplies about 130,000 frames with 12 AU intensities on a 0–5 ordinal scale [3]. The local archive contains 130,814 labeled frames. The original study trained intensity estimators under leave-one-subject-out cross-validation and reported ICC as a reliability measure. Our protocol is deliberately different: a general pretrained pipeline is evaluated without DISFA training, and F1 after a fixed presence threshold is primary. ICC(3,1) is retained as an exploratory intensity-consistency statistic [4].

III. METHODS

A. Deterministic Test Cohorts

For AFLFP, directory-name variants and spelling errors are mapped to the 16 canonical movement labels. A seeded

allocator selects at most one image for a subject-movement pair. With seed 42, the largest stable allocation satisfying both constraints contains 1,136 images: 71 per movement and 12 or 13 per subject. This prevents a long sequence from one subject or movement from dominating the result.

For DISFA, labels are first aligned from one-based annotation frame numbers to zero-based video indices. Seed 42 selects 200 frames uniformly without replacement inside each of the 27 subjects, yielding 5,400 frames. The left-camera videos are decoded sequentially and only selected frames are materialized as JPEGs. Sampling is subject-balanced, not class-balanced; therefore AU prevalence remains representative of the selected temporal frames.

B. Frozen Inference Protocol

Both datasets use Py-Feat 2.0.3 Detectorv2 with its distributed pretrained weights. Identity recognition is disabled. Images are passed through `detect()` with output size 512, batch size 32, zero data workers, and no augmentation. If more than one face is returned, the row with the highest finite face confidence is retained. There is no fitting, fine-tuning, calibration, validation split, or benchmark-driven threshold search.

Py-Feat inverse-maps the rescaled and padded Detectorv2 landmark coordinates to the original image frame before returning the result. We confirmed this path in the installed implementation and verified that the reported frame dimensions match the 1,280×720 AFLFP sources, avoiding a resized-prediction versus original-annotation coordinate mismatch.

Inference ran on an 8-core Apple M1 Pro with 16 GB memory, macOS 26.5.2, Python 3.12.12, PyTorch 2.13.0, and the MPS device. Wall-clock timing begins immediately before `detect()` and stops when it returns. It excludes package import, model construction, checkpoint access, DISFA ZIP extraction, video decoding, metric computation, and report generation. Consequently, FPS measures warm model throughput rather than end-to-end data ingestion.

C. Metrics

For a sample with $L = 68$ landmarks, prediction $\hat{\mathbf{p}}_i$, ground truth $\mathbf{p}_i$, and ground-truth bounding-box width w and height h , we compute

$$\text{NME} = \frac{1}{L} \sum_{i=1}^L \frac{\|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2}{\sqrt{wh}}. \quad (1)$$

NME is computed only for finite landmark predictions; missed faces remain in the detection-rate denominator. We report mean, population standard deviation, median, and 90th percentile across samples.

For AU a , DISFA intensity $y_a \geq 2$ is positive and Py-Feat score $\hat{y}_a \geq 0.5$ is positive. These thresholds were fixed before the final run. Per-AU F1 is

$$F1_a = \frac{2TP_a}{2TP_a + FP_a + FN_a}, \quad (2)$$

and macro F1 is the unweighted mean over 12 AUs. ICC(3,1) is computed between raw DISFA intensity (0–5) and raw Py-Feat score (0–1) using the fixed-rater, single-measure form

[4]. Because those scales are not calibrated, ICC is interpreted as an auxiliary consistency diagnostic rather than a calibrated intensity-estimation score.

D. Result Integrity

Seven dataset and metric tests pass, including annotation shape, frame-index alignment, balancing, missed-detection accounting, and threshold behavior. The paper generator independently recomputes every AFLFP summary and every DISFA F1/ICC value from sample-level CSV, failing on any discrepancy with the aggregate JSON. The final CSV contains exactly 1,136 AFLFP rows and 5,400 DISFA rows.

E. Target-Aligned Case Visualization

A separate seed-42 audit sample (256 AFLFP images and 270 DISFA frames) is used only to visualize outputs relevant to the planned oral-motor analysis. For AFLFP, we select the median-NME publication-eligible case for mouth closure, mouth opening, left smile, and right smile, using different subjects when available. For DISFA, we select different-subject cases with the highest manual intensity for AU12, AU25, and AU26, using Py-Feat probability as a tie-break, plus a target-inactive reference. The full selection keys, predictions, and landmark coordinates are retained in `target-case-manifest.json`. These cases illustrate the output format and do not replace the aggregate evaluation.

Two exploratory AFLFP geometries are shown with the landmark overlays. Mouth aperture is the distance between inner-lip landmarks 62 and 66, and lip-corner vertical difference is $|y_{48} - y_{54}|$; both are divided by the square root of the predicted 68-point bounding-box area. They are static-frame proxies for opening and asymmetry, not temporal clinical measures.

IV. RESULTS

A. AFLFP Landmark Localization

All 1,136 images yield finite 68-point predictions (100.0% detection). Mean NME is 6.368% with standard deviation 0.399%, median 6.361%, and P90 6.909% (Table I). Warm throughput is 24.96 FPS.

Figure 1 shows tight and similar movement distributions. Right wink is easiest at 6.271% mean NME, and tight eyes closure is hardest at 6.486%; the full mean range is only 0.215 percentage points. The descriptive comparison in Figure 1 places all Py-Feat movement means below the published Cascaded FCN means. This is encouraging external-generalization evidence, but it is not a controlled superiority claim because the training sources and evaluated cohorts differ.

Figure 2 connects the landmark benchmark to the intended oral-motor measurements. The selected open-smile case has 5.66% normalized aperture, compared with 0.29% in the close-smile case. NME for the four displayed cases ranges from 6.26% to 7.44%. The left/right cases also show that a single-frame lip-corner height difference is visible but is pose-sensitive and should be confirmed over a sequence.

TABLE I

HEADLINE TEST-ONLY RESULTS. ACCURACY METRICS ARE TASK-SPECIFIC AND MUST NOT BE COMPARED ACROSS ROWS. THROUGHPUT EXCLUDES MODEL INITIALIZATION AND DISFA FRAME DECODING.

Dataset	Task	Test samples	Detection	Accuracy	Throughput
AFLFP	68-point landmarks	1,136	100.0%	NME 6.368%	24.96 FPS
DISFA	12-AU detection	5,400	100.0%	macro F1 0.714	26.14 FPS

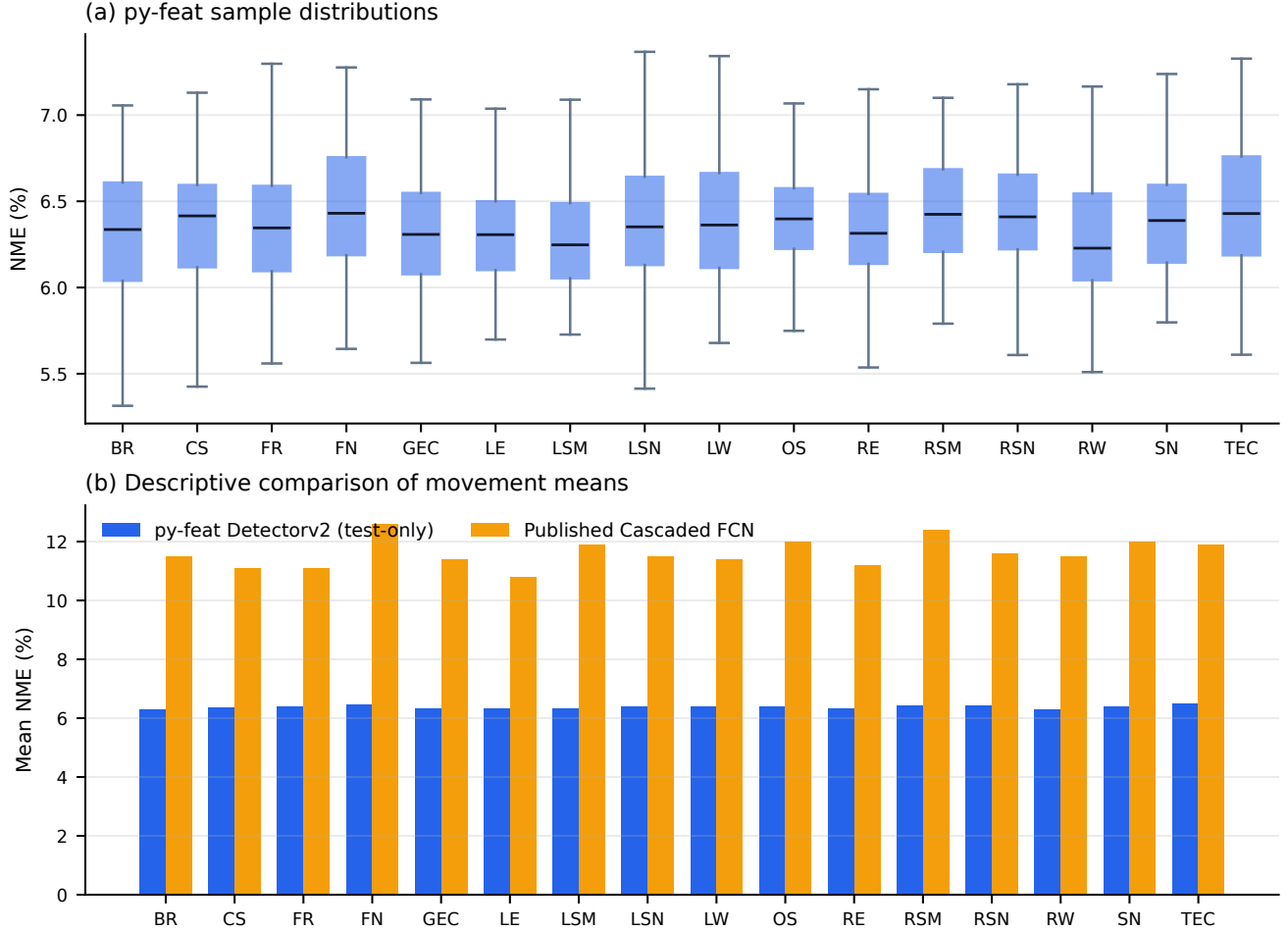


Fig. 1. AFLFP landmark localization by movement. (a) Boxplots summarize the 71 test-only samples in each movement; whiskers follow the standard 1.5-IQR rule and outliers are hidden for legibility. (b) Cohort means are shown beside the Cascaded FCN means reported by Xia et al. [2]. The latter are descriptive references, not matched-split comparisons.

B. DISFA Action Units

All 5,400 selected frames yield finite AU predictions (100.0% detection). Macro F1 is 0.714 and macro ICC(3,1) is 0.467, with warm throughput of 26.14 FPS. AU performance is heterogeneous (Figure 3 and Table III). The strongest F1 values occur for AU25 (0.959), AU02 (0.917), and AU04 (0.878). The weakest occur for AU20 (0.479), AU17 (0.507), and AU05 (0.548).

Prediction counts expose distinct error modes. AU05 has 59 positives but only 25 predicted positives, and AU20 has 132 versus 85, consistent with under-detection. Conversely, AU17 has 282 positives but 780 predicted positives, and AU12 has 671 versus 1,123, consistent with over-detection at the fixed threshold. These count asymmetries explain why a single

macro score is insufficient.

Across the 12 AUs, positive prevalence and F1 have a descriptive Pearson correlation of $r = 0.708$. The pattern is compatible with more stable measurement when more positives are available, but it does not establish a causal effect: AU identity, visual distinctiveness, co-activation, and model training frequency are confounded. AU05, for example, has only 59 positive frames (1.09%), whereas AU25 has 1,497 (27.72%).

Figure 4 shows the target outputs on actual DISFA frames. The AU12 case has manual intensity 5 and Py-Feat probability 1.00; the AU25 case has 5 and 1.00; and the strongest sampled AU26 case has 3 and 0.99. The target-inactive reference produces 0.00 for all three probabilities. Co-activation is visible in the positive cases, reinforcing that AU12, AU25, and AU26

TABLE II
AFLFP NME BY MOVEMENT (%). POPULATION SD AND P90 ARE COMPUTED FROM THE SAVED PY-FEAT SAMPLE PREDICTIONS. THE FINAL COLUMN IS TRANSCRIBED FROM THE PUBLISHED CASCADED FCN ROW [2].

Movement	Abbr.	<i>n</i>	Mean	SD	Median	P90	Cascaded FCN mean
brow raise	BR	71	6.296	0.393	6.336	6.717	11.5
close smile	CS	71	6.350	0.395	6.415	6.802	11.1
frown	FR	71	6.372	0.408	6.346	6.944	11.1
funny	FN	71	6.448	0.398	6.431	6.937	12.6
gentle eyes closure	GEC	71	6.312	0.373	6.308	6.755	11.4
left eyebrow	LE	71	6.317	0.344	6.306	6.799	10.8
left smile	LSM	71	6.310	0.350	6.247	6.773	11.9
left snarl	LSN	71	6.378	0.447	6.352	6.963	11.5
left wink	LW	71	6.392	0.391	6.362	6.886	11.4
open smile	OS	71	6.396	0.305	6.398	6.764	12.0
right eyebrow	RE	71	6.315	0.355	6.314	6.726	11.2
right smile	RSM	71	6.429	0.404	6.424	6.916	12.4
right snarl	RSN	71	6.416	0.402	6.410	6.842	11.6
right wink	RW	71	6.271	0.418	6.228	6.849	11.5
snarl	SN	71	6.398	0.472	6.388	6.998	12.0
tight eyes closure	TEC	71	6.486	0.423	6.429	7.011	11.9

TABLE III
DISFA TEST-ONLY RESULTS BY AU. POSITIVE SUPPORT USES INTENSITY ≥ 2 ; PREDICTED POSITIVES USE SCORE ≥ 0.5 . EACH AU IS EVALUATED ON 5,400 FINITE PREDICTION/LABEL PAIRS.

AU	Positive support	Prevalence (%)	Predicted positive	F1	ICC(3,1)
AU01	299	5.54	389	0.695	0.409
AU02	261	4.83	245	0.917	0.472
AU04	800	14.81	934	0.878	0.469
AU05	59	1.09	25	0.548	0.318
AU06	409	7.57	590	0.763	0.539
AU09	208	3.85	237	0.715	0.442
AU12	671	12.43	1,123	0.737	0.536
AU15	100	1.85	198	0.591	0.527
AU17	282	5.22	780	0.507	0.504
AU20	132	2.44	85	0.479	0.331
AU25	1,497	27.72	1,606	0.959	0.528
AU26	440	8.15	571	0.785	0.525

should be analyzed jointly rather than as mutually exclusive labels.

V. DISCUSSION

The landmark result is the clearest evidence of cross-domain utility. Detectorv2 detects every selected AFLFP face and keeps movement means within a narrow interval, despite facial asymmetry and movement-specific deformation. Its 6.368% mean NME is also substantially below the 11.6% historical Cascaded FCN reference. The comparison should motivate a matched, full-corpus replication, not a leaderboard claim: the older baseline was trained and evaluated under a different protocol.

The DISFA result is more nuanced. Macro F1 0.714 indicates useful zero-shot AU presence detection, but the 0.480 range between the best and worst AUs is large. Prevalence and predicted-positive counts show both data scarcity and threshold miscalibration. A deployment that treats every AU score identically would therefore conceal predictable failure modes. Per-AU calibration on a separate validation population could improve a downstream application, but doing so would no longer be the test-only setting evaluated here.

The macro ICC of 0.467 should not be read as a direct comparison with DISFA-trained intensity regressors. Py-Feat scores are bounded to 0–1 while DISFA intensity spans 0–5, and no scale mapping was learned. The statistic still reveals ordinal consistency differences across AUs, but an intensity study should reserve subjects for calibration, map scores to the DISFA scale, and report the mapping separately.

The case visualizations make the project linkage concrete. AFLFP supplies observable mouth-opening and lateral-movement geometry, while DISFA supplies AU12/25/26 activation patterns on actual faces. The current pipeline therefore provides candidate inputs for oral-opening, closure, and lower-face movement analysis. It does not yet measure lip speed, periodicity, audio–visual synchrony, or task-level clinical discrimination; those require continuous Task 2/5 recordings and participant-level validation.

Warm throughput near 25–26 FPS demonstrates that batched offline processing is practical on the measured Apple M1 Pro. It does not imply real-time end-to-end video throughput: camera capture, decoding, transfer, UI rendering, model initialization, and result serialization are explicitly excluded.

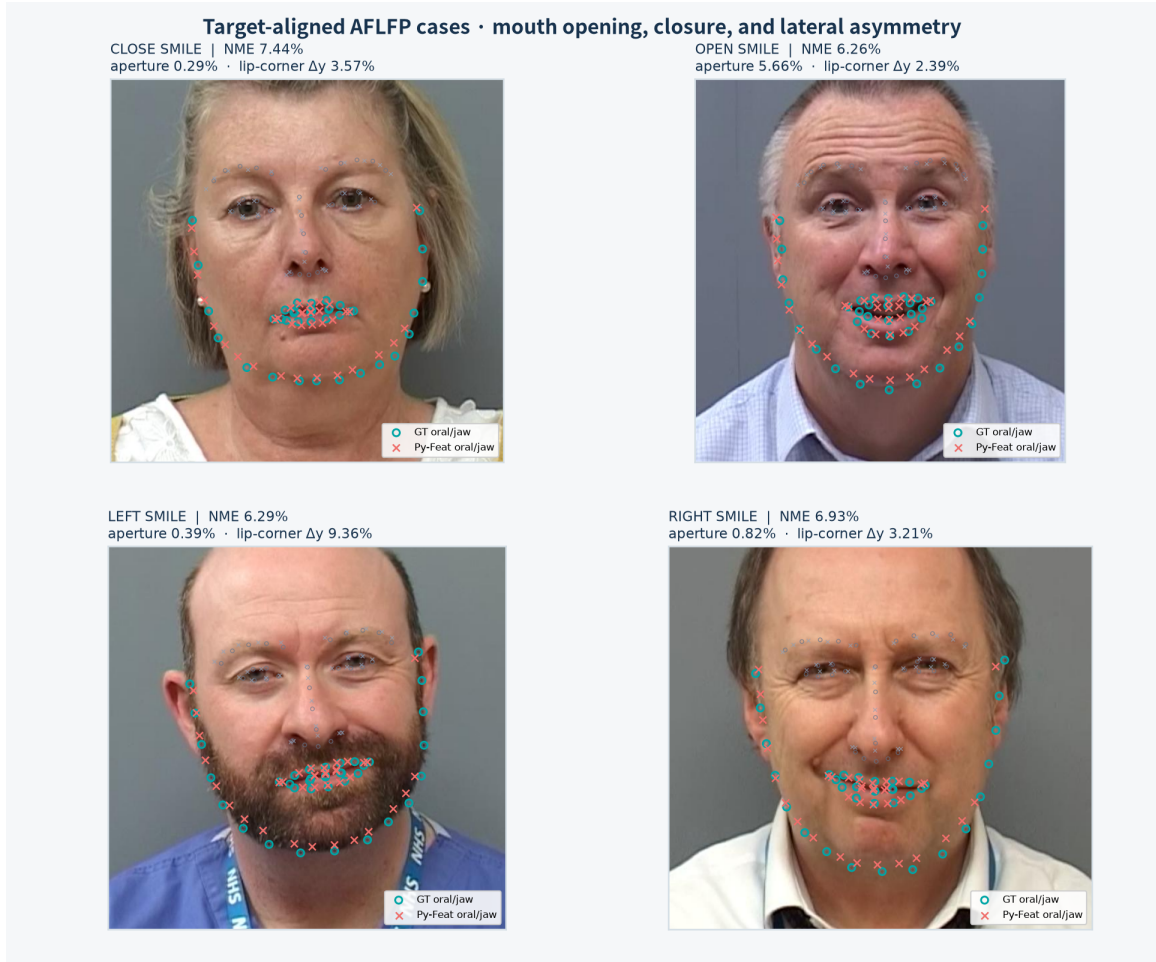


Fig. 2. Target-aligned AFLFP landmark cases. Gray marks show all 68 points; teal circles and coral crosses emphasize ground-truth and Py-Feat oral/jaw points. The panels cover mouth closure, mouth opening, and left/right lip-corner movement. NME and exploratory normalized oral geometries are shown for each publication-eligible case.

VI. LIMITATIONS

First, both cohorts are deterministic samples rather than full-corpus evaluations. AFLFP covers 1,136 of 5,328 paired annotations and DISFA covers 5,400 of 130,814 labeled frames. Subject and target coverage reduce sampling bias but cannot remove it; one seed also does not quantify sampling variability. Second, the AFLFP released file inventory differs from the image count described in the source paper, so the present result is tied to the inspected local distribution.

Third, test-only execution prevents local leakage through fitting or tuning but cannot prove that upstream model-development data were disjoint from these benchmarks. Fourth, the DISFA protocol evaluates independent frames from the left camera and ignores temporal structure. JPEG materialization may also cause small score changes near the 0.5 threshold. Fifth, AU F1 binarizes an ordinal task and ICC compares unmatched score ranges. Finally, AFLFP performance is not clinical validation; landmark accuracy alone does not establish safety or diagnostic utility for facial palsy. The four target-aligned cases per dataset are selected to expose the intended signals, not to represent their population distributions, and their static geometry is sensitive to head pose.

VII. REPRODUCIBILITY AND DATA GOVERNANCE

The repository retains benchmark commands, aggregate JSON, sample-level CSV, generated table fragments, vector figures, software versions, and this LaTeX source. Running `make pdf` validates the saved outputs, regenerates the aggregate derived assets, and compiles the manuscript. Raw AFLFP and DISFA media remain excluded because access and redistribution are governed by their respective dataset terms. The manuscript includes only the selected publication figures: AFLFP cases are restricted to subjects permitted for academic publication, and DISFA is shown as target-focused face-region crops. The target-case manifest stores selection keys, outputs, and coordinates but no raw media. No clinical or hospital participant data were used, and no new human-participant data were collected.

VIII. CONCLUSION

Under a frozen, test-only protocol, Py-Feat Detectorv2 provides complete face detection and consistent landmark localization on the sampled AFLFP cohort, with mean NME 6.368%. On subject-balanced DISFA frames it reaches macro F1 0.714, but AU-specific results vary widely and raw intensity

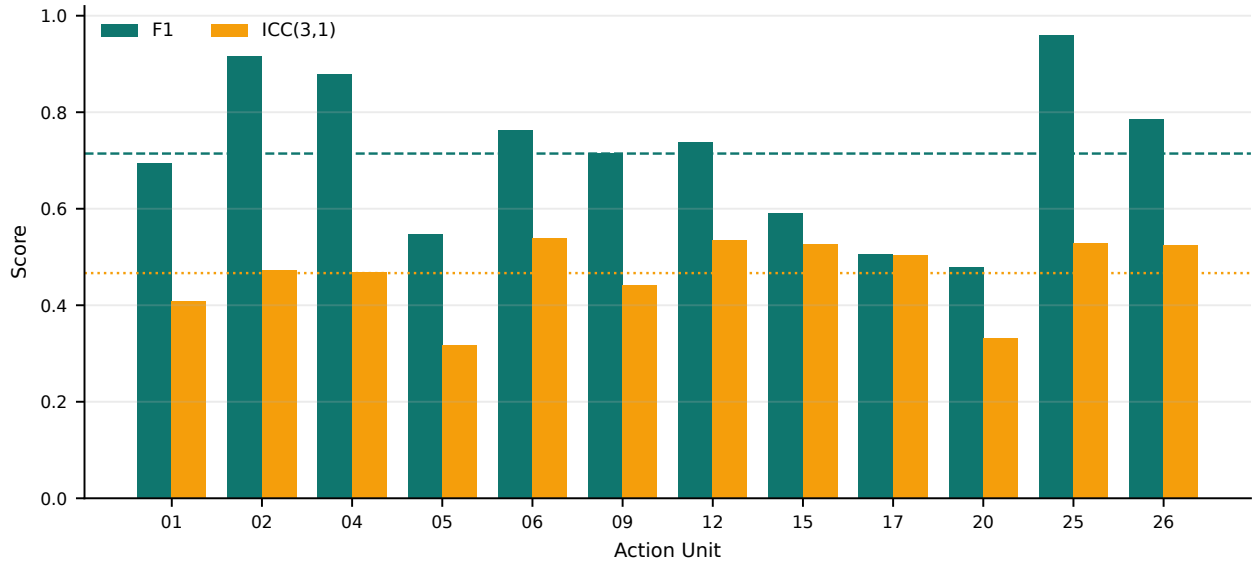


Fig. 3. DISFA per-AU F1 and ICC(3,1). Dashed and dotted horizontal lines are the respective macro means. F1 is the primary presence metric; ICC is auxiliary because the 0–5 ground-truth intensity and 0–1 Py-Feat output are not calibrated to a common scale.

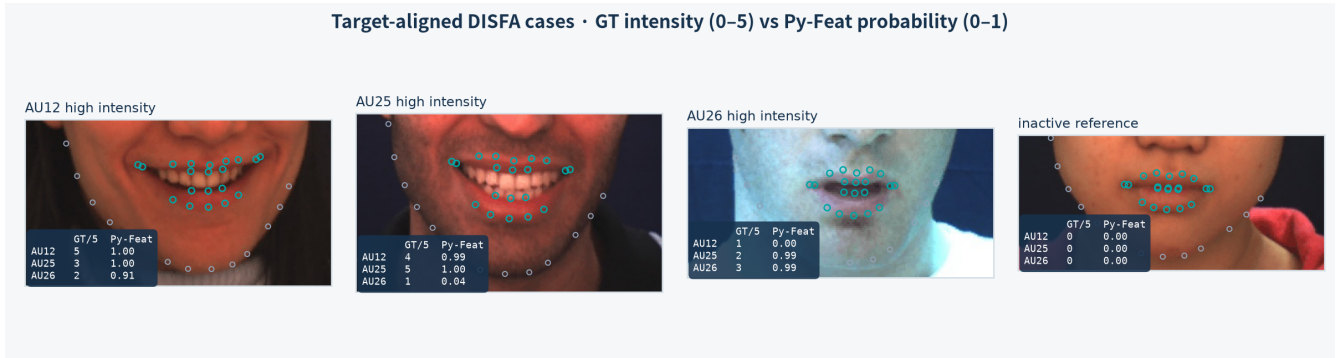


Fig. 4. Target-aligned DISFA cases for AU12 (lip-corner pull), AU25 (lips part), AU26 (jaw drop), and a target-inactive reference. Teal points are the Py-Feat mouth landmarks and gray points trace the jaw. The inset compares manual intensity (GT, 0–5) with Py-Feat probability (0–1); the scales are not calibrated. These deliberately clear high-intensity examples are qualitative and are not estimates of cohort-level performance.

consistency remains limited. The practical conclusion is task-dependent: Py-Feat is a strong external landmark baseline here, while AU deployment requires per-class inspection and, when permitted, independent calibration. The released result chain makes both findings directly auditable.

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